

Investigating the Correlation Between Presence and Reaction Time in Mixed Reality

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Measuring *presence* is critical to improving user involvement and performance in Mixed Reality (MR). However, current methods utilize questionnaires that do not capture time-varying responses and are prone to user bias. Inspired by the existing literature on the relationship between *presence* and human performance, we argue that there is a potential to leverage existing user interactions in the virtual scene as a measure of *presence*. We explore the user *reaction time* as one such quantity that can be easily measured using the systemic tools available in modern MR devices. We conducted an exploratory study (N=40) and found a significant correlation between *presence* scores and *reaction times* with a correlation coefficient of -0.65. While our study points to an inverse relationship between *presence* and *reaction time*, we also note that the slope of the negative correlation may depend on the scenario under consideration and needs further investigation.

1 INTRODUCTION

After the popularity of Virtual Reality (VR) and Augmented Reality (AR), Mixed Reality (MR) is gaining importance in science, education, training, and entertainment, affording new ways of interaction and engagement with the real and virtual worlds. Recent advances in tools and technology for MR have made it possible for a user to feel “present” in the mixed reality environment. *Presence* is typically described as the subjective experience of being in a simulated place or environment, despite knowing that you are physically located in a different real environment [6, 27]. A high *presence* is desired, as it allows the user to experience an immersive, realistic, and involved experience in the virtual environment.

There are many studies that investigate the notion of *presence* and describe the factors that contribute to the sense of *presence*. According to Witmer & Singer [54], Slater & Steed [46] and others [6, 27, 36, 37], there are two main aspects of *presence*: *place illusion* and *plausibility illusion*. *Place illusion* refers to the *sense of being there*. In an MR environment, this corresponds to the extent to which the virtual content appears to be indistinguishable from the real world. The second factor is *plausibility illusion*, which refers to the sensation that the observed events depicted in a virtual environment actually occur. When both *place illusion* and *plausibility illusion* occur, users will feel involved in the environment and will respond realistically to the environment, leading users to experience greater engagement in a mixed reality environment. Both immersion and participation are necessary to experience *presence* [54].

A prerequisite to improving *presence* of a user is the ability to measure and quantify *presence*. Due to its subjective nature, the most popular method to measure *presence* is the use of subjective questionnaires [12, 22, 38, 39, 54]. Questionnaires ask users to self-report their sense of *presence* answering questions that attempt to assess *presence*, usually after the user has left the virtual environment. The benefit of subjective questionnaires is that they can be quickly administered, graded, and interpreted without affecting the user experience. However, the questionnaire cannot measure the time-varying qualities of *presence* and can produce unstable, inconsistent, and irreproducible responses due to the prior experience of the participants [8]. The well-known shortcomings of *presence* questionnaires have led researchers

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to explore alternative approaches to assessing presence, including behavioral responses such as postural response[9], hand and eye response [51], startle response [53], which measure participant responses that are produced automatically, without conscious thought, thus avoiding user bias. However, they are susceptible to experimenter bias and are highly sensitive to environmental factors and content [20, 47]. There are also various physiological responses that can be measured to assess *presence*, such as a change in heart rate [11, 25], skin conductance, and body temperature [19]. However, physiological measures can also be noisy and unreliable, especially under non-stationary conditions, and may not capture differences in presence in situations of low emotional valence [30].

Given the state-of-the-art, there is a need for an approach to quantifying *presence* that is objective, discrete, not consciously affected by the participant and/or experimenter, and could be used at runtime without interfering with the virtual experience. It should take advantage of existing interactions (and underline quantities) that are happening in the virtual scene and measure *presence* through these interactions rather than making additional external interventions. One of such underlying quantities is *reaction time*. *Reaction time* or response time refers to the time taken between when humans perceive something and when humans respond to it. *Reaction time* is dictated by the cognitive ability to detect, process and respond to a stimulus [5, 52], and can be easily measured using the systemic tools available in modern MR devices such as Microsoft HoloLens 2 [15].

Our work investigates a fundamental question in mixed reality: Would an individual experiencing more *presence* systematically show faster *reaction times*? If the answer is yes, we could use a systemic metric such as *reaction time* to quantify *presence* in a non-intrusive, objective, and unbiased manner. There is a large body of work investigating the relationship between *presence* and human performance [3, 21, 23, 28, 29, 31]. Natalia et al. showed a negative correlation between task completion time and *presence* when the sense of *presence* is altered by multisensory feedback [3]. Matteo et al. show a correlation between performance and *presence* when the sense of *presence* is changed by varying the perceptual load [29]. Maneuvrier et al. showed that presence promoted spatial cognition performance and that the presence-performance relationship was not mediated by other human factors [28]. Furthermore, human performance is often used as an argument for the good predictive validity of questionnaires [12].

To understand the relationship between *presence* and *reaction time*, we conducted a study in which we change *presence* of users by manipulating *place illusion* and *plausibility illusion* when interacting with an MR environment. We designed two sets of experiments. In one set, we only manipulated the appearance of the virtual object and, in the other set, we manipulated a non-task-relevant behavior of the virtual object. All other aspects of the experiments, such as the interaction mechanism, frequency of interactions, and physical environments, were kept the same. We systemically measure the *reaction time* of users in response to a visual stimulus. Our post-experience questionnaires show a significant change in the presence between experiments. Similarly, we see a significant change in user *reaction time* as the sense of *presence* changes across experiments. Our analysis shows a correlation between *presence* and *reaction time*.

In our attempt to understand the *presence* and *reaction time* relationship, we make the following contributions.

Contribution 1: We propose the use of *reaction time* as a measure of *presence*. We also develop a non-intrusive, systemic approach to measuring the user *reaction time* that relies on existing interactions in MR environments.

Contribution 2: We devise experiments that alter the sense of *presence* by manipulating *place illusion* and *plausibility-illusion*. In designing experiments, we control for other factors that are known to have an impact on user performance.

Contribution 3: We conduct an exploratory lab study ($N = 40$) with our setup, demonstrating a negative correlation between the sense of *presence* and the user *reaction time* when responding to a visual stimulus.

2 BACKGROUND

2.1 Factors affecting presence

Presence is a phenomenon of awareness based on the interaction between sensory stimulation, environmental factors that encourage involvement and allow immersion, and internal tendencies to become involved [54]. Place illusion ('sense of being there') can be affected by the realism of the virtual content, consistency of the view between headset and direct view from the users' eyes (including displacement or latency issues), lack of haptic feedback, and being overly aware of the headset. The second factor, plausibility illusion, can be affected by scene elements not obeying the laws of physics, cause-effect relationships not coupled as expected, actions that do not have the expected outcome, and events in the virtual environment not conforming to familiar expectations. It is argued that when both place illusion and plausibility illusion occur, users will feel involved in the environment and respond realistically to the environment, which will lead users to experience greater engagement in a virtual environment. Both involvement and immersion are necessary to experience *presence* [54].

2.2 Presence questionnaire

The most commonly accepted method of evaluating *presence* is the self-report questionnaire [4]. The presence questionnaires are used after participants engage in an MR environment (a post-experience questionnaire). The Witmer & Singer Presence Questionnaire (PQ) is one of the most widely used [54]. Other popular post-experience questionnaires include the Igroup Presence Questionnaire (IPQ) [38, 39] and the Slater-Usch-Steed Questionnaire (SUS) [46]. The PQ questions are based on the following factors: ability to control the environment and "naturalness" of control over the environment; coherence, consistency of information arriving from different senses, and distractions a participant may experience in a virtual environment; environment realism and meaningfulness; disorientation when returning to the real world. The SUS and IPQ questions are based on three factors: the sense of physically being in the virtual environment, the extent to which the virtual environment feels real, and the extent to which the participant feels involved in the environment.

2.3 Taxonomy of the virtual scene

As described by Wilbur and Slater [54], the place illusion (appearance realism) and the plausibility illusion (behavioral realism) are the main aspect of any virtual experience in any MR medium.

- **Virtual scene** represents the semantics of the virtual environment in three dimensions (3D) placed on top of the real environment.
- **Event** is an action or occurrence recognized by the computer system. For example, when a virtual object is added or removed from the virtual scene, this can be dubbed as an event.
- **Task** refers to an activity that is observable and has a start and an end point. In MR, tasks will be aligned with the start or end of an event, depending on the semantics of the virtual scene.
- **Interaction** is defined as the act of performing a physical action to perform the task in a virtual scene.
- **Cue** is defined as the signal (visual or auditory) that is sent to the participant to initiate a task.
- **Feedback** is defined as the visual or auditory confirmation sent to the participant that the task is completed.

To create a virtual scene that has context, immersion, and interaction, we need to craft our experiments so that participants feel engaged in the virtual world. However, the scene should neither be too complex that the participants' cognitive load is consumed in understanding the scene, nor should the scene be too simple that the participant feels disengaged. For example, we do not want a scene with Warcraft heavy-load games, nor do we want a simple box that has no semantic value to the participant. We need to create a balance between a semantically too complex and a simple virtual scene. The same applies to the events, tasks, cues, interaction, and feedback that arise from this virtual scene.

3 STUDY DESIGN

To examine the validity of *reaction time* as a measure of *presence*, we investigate *correlation between presence and reaction time* in mixed reality (MR). For any measure to be useful, Insko et al. [36] suggest the following criteria (C).

- **C1: Sensitivity** - a measure should be able to distinguish between various feelings of *presence*. To test this, we need to create varying feelings of *presence*.
- **C2: Reliability** - a measure should produce repeatable results across trials and across participants.
- **C3: Validity** - a measure should be able to quantify subjective *presence*, or at least correlate well with established subjective *presence* measures. We need to establish a baseline from existing questionnaire based measures and demonstrate correlation with those measures.
- **C4: Objectivity** - a measure should be free from participant and experimenter bias. The tasks and scenarios that we create should not offer an advantage to a certain group of users over others. Tasks should also not be deliberately designed to push the results in the favor of the experimenters' hypotheses.

To meet the above mentioned criteria, we face the following design challenges (DC).

- **DC1:** How can we induce different feelings of *presence*? (C1: sensitivity)
- **DC2:** How do we *minimize external variables* while varying feelings of *presence*? (C2: reliability)
- **DC3:** How do we *establish a baseline* and what that baseline should be? (C3: validity)
- **DC4:** What should the *interaction mechanism* be? (C4: objectivity).

3.1 DC1 - Varying presence

For measuring varying feelings of presence, we need an empirical setup that can measure presence in real-time while depicting a practical scenario. But first, we need to understand what the main aspects of presence are and what dimensions could be measured in those distinct but overlapping aspects. We describe two main aspects (place illusion and plausibility illusion) of presence in Section 2. Since we are using Mixed Reality (MR) as our medium of experiment, place illusion and plausibility illusion would need to be refined. In MR measuring the place illusion could mean to what extent the virtual object is appearing to be indistinguishable from reality. The sense of plausibility can be described when users select a dominant space as the reference frame. Then virtual objects in real space or real objects in a virtual space would be perceived as plausible if the object behaves coherently to the dominantly perceived space as noted in place illusion. For example, gravitational forces are applied horizontally rather than vertically.

In summary, place illusion mainly refers to the elements related to *appearance* of the environment, whereas plausibility illusion refers to the elements related to *behavior* of the objects in the environment. Therefore, we argue that to vary the feelings of presence, we could manipulate *appearance* and/or manipulate the *behavior* of the object in the scene.

3.2 DC2 - Controlling confounding variables

There are a variety of options for manipulating the appearance and behavior of objects in the virtual environment. To satisfy the reliability criteria, we need to carefully pick scenes and manipulations. We have identified two constraints that will help limit the introduction of external variables and maintain symmetry across experiments and users.

- Constraint 1: Manipulations should not affect (increase or decrease) the overall complexity of the scene, including tasks, events, and interactions. (**simplicity** in scene)
- Constraint 2: Manipulations should also be free from additional confounding variables. Confounding variables are the extra variables that affect the actual relationship between the variables under study [32]. In our case, these two output variables are *presence* and *reaction time*. (**symmetry** in the scene and users)

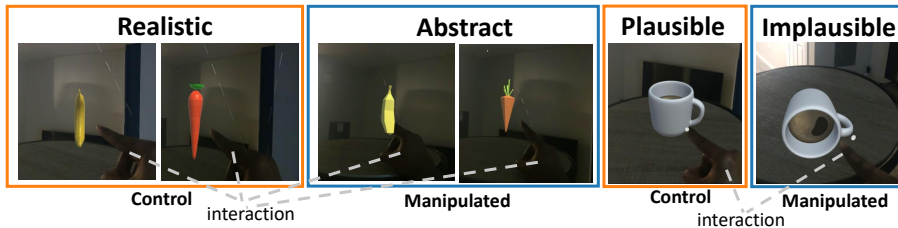


Fig. 1. Realistic, abstract, plausible, and implausible virtual objects used in the experiments.

To satisfy the simplicity constraint, we will avoid unnecessary details in the scene (extra buttons, features) and keep experimental conditions the same across all trials and users. These experimental conditions are the placement and duration of the visual stimulus, time and appearance of the cue, time and appearance of the feedback, number of tasks, duration of the experiment, and type of interaction. We also argue that to keep the scene simple, we should pick a scene that is familiar to most college students (intended participants pool) and have only one virtual object in the scene at a time. However, we want to keep the participant engaged throughout the experiment, and we need a scene with some familiar semantics. Similarly, the manipulations that will vary the presence should be subtle and change only one entity of the scene to isolate the effect on our output variables. In [41] authors have proposed manipulating the appearances of the scene by manipulating the visual fidelity. We can measure the varied presence by making the scene appear *realistic* in one scenario and manipulate the same scene by making its appearance *abstract*. Similarly, as demonstrated in [1], plausibility illusion can be affected if the laws of physics are challenged in a scene. In behavior manipulation, we can make the behavior of a scene naturally plausible in one scenario and implausible in another.

To satisfy the symmetry constraint, we will maintain a controlled physical environment to minimize interference from the physical world. This includes keeping the same room, lighting, seating arrangement, duration of the study session, breaks, order of the experiment among all users, and experiments.

To this end, we divide our experiment into two sets. In the first set, we vary the sense of presence by manipulating the *appearance* of the virtual object (place illusion). In the second set, we vary the sense of presence by manipulating the *behavior* of the virtual object (plausibility illusion). Each experiment will contain two blocks of trials: a *control* block and a *manipulated* block. In the *control* block, the scene contains virtual objects that have a natural appearance and plausible behavior. However, in the *manipulated* block, the scene contains virtual objects whose appearance is manipulated to be visually unnatural (unrealistic) or in which the virtual objects exhibit implausible behavior.

Experiment Set 1: Realistic vs. Abstract. In the *control* trials, all virtual objects in the scene will have a realistic appearance (textured and natural) and plausible behavior. We refer to this block of trials as <realistic> in the rest of the paper. In the *manipulated* trials, all of the virtual objects will depict plausible natural behavior, but the appearance would be abstract (untextured and geometric). We refer to this block of trials as <abstract> in the remainder of the paper.

For *realistic* versus *abstract*, we need a scene that has textured virtual objects. To satisfy the simplicity constraint, we have modified the popular Fruit Ninja game [48] (more than 1 billion downloads and 2 million active players worldwide daily). To mimic the semantics of a multi-object environment, we needed at least two different objects that are similar in shape and size and different in textural properties. Since we want a simple scene and only one virtual object in a scene at a time, virtual objects will appear in the scene one after the other. In the end, we picked banana (fruit) and carrot (vegetable) for this experiment. We also removed complex interactions (slicing) and cognitive loads (scores) from the original game. In the *realistic* scene, we made the carrots and bananas appear as natural as possible in terms of their

color, texture, and shape. In the *abstract* scene, we rendered the carrots and bananas with muted colors, without texture, and geometrically blocky shapes, as shown in Figure 1.

Experiment Set 2: Plausible vs. Implausible. In the *control* condition, the virtual objects in the scene have a natural appearance and plausible behavior. We will refer to this block of trials as <plausible> in the remainder of the paper. In *manipulated* condition, the virtual objects retain a natural appearance, but behave implausibly from real-world standards (disobey laws of physics). We will refer to this block of trials as <implausible> in the remainder of the paper.

For the *plausible* vs. *implausible* scenario, we chose a scene with elements that have physical constraints. We modified the popular Coffee Stack game [10] with a coffee mug, on a surface that periodically fills up with coffee. Like Fruit Ninja, we have also removed complex interactions and cognitive loads from this game. In the *plausible* scenario, the empty white porcelain coffee mug will be upright and the coffee will continue to pour into the mug periodically until the mug is full. Then, a new mug will appear, and this cycle will continue to repeat. In the *implausible* scenario, the empty white porcelain coffee mug is in a tilted position on the surface, and coffee continues to pour into and fill the mug even though gravity and volumetric constraints should not allow this (see Figure 1).

3.3 DC3 - Establishing a baseline

Since the presence itself has a subjective nature, it is logical that we also establish our baseline with a subjective measure. We establish the baseline using a subjective questionnaire, as its use is an accepted standard measure of presence in the research community [1, 34, 41]. Subjective questionnaires have been the standard measure of presence for many years and have been shown to be sensitive enough to find differences in presence when used to examine the difference between two visually similar fidelities [36]. The post-experience questionnaire provides scores that reflect the level of perceived presence in the scene. We measure various feelings of presence with the three most-cited and common questionnaires that measure the main aspects of presence. We use the Igroup Presence Questionnaire (IPQ - cited 758 times) [38, 39], Slater-Usoh-Steed (SUS - cited 853 times) [46] and Witmer & Singer (WS - cited 3596 times) [54].

3.4 DC4 - Selecting an interaction mechanism

We need a task that engages the participant and helps us measure our output variable, *reaction time*. Through interactions, the participant can physically be involved in the scene. In designing the interaction, we want to avoid imposing differential barriers to task completion between participants. An example of such effort would be to place a push button at an accessible height for all the participants.

Similarly, we want to avoid designing unnecessarily complex interactions that could potentially cause task performance to vary in unpredictable ways, independent of the targeted manipulations. Additionally, as part of our design, we ensure that the reaction time is not affected by external interventions. Therefore, we need to leverage existing elements in the taxonomy of the scene (cue, interactions, and feedback).

For this reason, we have employed a simple 'tap' as our interaction mechanism. It is precise and does not respond to direction and / or length of movement or any other physical attribute of a typical participant.

We designed a task that supports participants in knowing what to do (cue), knowing that the system is working (interaction/tap), and knowing if their action was understood by the system (feedback). In the *realistic* vs. *abstract* scenario, the appearance of the object in the scene is a cue for the user to take an action (tap) and the object leaves the scene as a feedback for a successful tap. In *plausible* vs. *implausible*, the change in coffee level is the cue to take an action (tap) and the cup changes color as a feedback for successful tap (more details in Section 4.4).

Next, we formally define the reaction time ($reaction_i$) in our study ($reaction_i$) for the trial i as follows:

$$reaction_i = interaction_i - cue_i$$

where trial i refers to one task iteration by the participant (appearance, tap, feedback). $interaction_i$ is the time when the tap is recorded in the trial i and cue_i is the time of onset of the task cue in the trial i .

4 USER STUDY

In this section, we will provide details on the study measures, participants, equipment, and procedure for the user study.

4.1 Measures

Presence scores for the questionnaires were obtained using a total of 39 items (6 SUS, 14 IPQ, 19 WS) on a 7-point scale. We did not modify any of the questions. The reaction time is recorded by our software on HoloLens 2.

4.2 Participants

We recruited a total of 40 participants (23 male identified, 16 female identified, 1 non-binary identified) with a mean age of 26.6 years (standard deviation of 5.5 years). All participants volunteered and provided an in-writing informed consent. They received \$25 for their participation. All but one participant had a technical background in computer science or engineering. All the participants had normal or corrected normal vision with contact lenses or glasses. 17.5% of the participants had 1 to 4 days per week MR experience, 24 participants had less than 1 hour per week experiences, and 9 had never experienced MR before. Only 12 participants had used HoloLens 2 before the study. The study was granted ethics clearance according to the ethics and privacy regulations of our Institutional Review Board (IRB).

4.3 Equipment

The virtual environment for the study was presented on an ergonomic, untethered, self-contained holographic device, HoloLens 2 [15]. It has a second-generation custom-built Holographic Processing Unit (HPU) which enables real-time computer vision with enterprise-ready applications powered by a Qualcomm Snapdragon 850 CPU that runs all applications. All virtual scene applications were developed using the Unity 3D game development engine (2020.3.26f1) with API for the Universal Windows Platform on Windows 10 PC configured with SDK 10.0.19362.0.

HoloLens 2 is configured with the eye, spatial, and hand tracking input systems, along with the camera and other sensors with 43° horizontal 29° vertical 52° diagonal field of view. HoloLens 2 has two optical see-through displays [24] with a screen resolution of 1440x936 pixels (one for each eye) and a refresh rate of 60Hz. It uses a tinted visor to reduce the amount of light from the environment that reaches the user's eyes.

4.4 Pilot Study

Before starting the formal study, we conducted a pilot study with two participants. to tune the parameters to minimize environmental variables and maximize reliability and objectivity. We use a talk-aloud protocol and ask participants questions to align the experiment for general comfort and ease for the participants, but not to bias the experiment for a specific set of users. For session-specific parameters, we tuned the following parameters.

Participant position. We experimented with standing, walking, and sitting positions. The participants reported feeling tired while standing. While walking, they reported that the scene kept changing around them, necessitating extra attention to locate the virtual object. Participants reported the sitting position as the most comfortable position for the experiment. We asked the participant to sit on a chair with relaxed shoulders, an arm on the lap or armrest, and feet flat on the floor. The chair with armrest and backrest was reported to be the most comfortable with air-tap interaction [14].

Room lighting. Lighting affects a user's ability to see the virtual environment and its objects [7]. It also affects the rendering of virtual objects and user interaction with those objects, as they rely on the tracking module of HoloLens 2. For optimal visuals through HoloLens 2, lighting should be even and sufficiently bright so that a participant can see without effort, but not so bright so that a participant has a hard time looking into the environment. To compensate for

the loss of the visor dim lights reflecting in the direction of the participant's head are deemed most effective because a tinted visor may cause a loss of contrast in the physical environment.

Experiment duration. How long will an experiment run with the same repetitive task? It should be long enough for the participant to feel involved and we can collect a sufficient amount of data on *reaction time*. It should be short enough that the participant does not feel tired or disengaged impact the accuracy of their interactions. We tested a duration of 2 to 10 minutes. At 2 minutes, the participants reported that they were unable to get acquainted with the environment. At 10 minutes, the participants reported feeling disinterested after a while. We also wanted to test the recovery time of the presence or reaction time (discussed in detail in 5.2). We picked 5 minutes per experiment as it allowed us to obtain at least 60 reaction time readings per experiment and the participant did not feel tired.

Wait time between blocks and experiments. After each block of an experiment, the participant was asked to complete the questionnaires. We explored 0-20 minutes of wait times after filling out the questionnaire and before starting the next block. Participants reported 5-minute wait times as sufficient, but a longer break was available upon request. In the main study, we explicitly asked participants before each experiment if they needed more downtime.

Experiment and block order. For an experiment, we could expose the participant to a control block and then to manipulated trials, or vice versa. In the latter case, there is a risk of obtaining better performance for the control block as a result of long exposure. In the former case, the user will get familiar with the experiment and its performance should improve for the manipulated block (a bad result for us). In our pilot study, we ran the first order with one participant and the second order with the other participant. We did not observe any difference between the two participants and decided to run all participants with control block first and manipulated block second. We agree that potential order effects warrant further investigation in the future, but we note that the ordering we chose is probably least likely to disadvantage task performance in the manipulated trials, due to the longer duration of prior practice.

Virtual object placement. We use the Fitts's law [13] to calculate the expected time of motor movement for several different positions in the scene. We decided to place the virtual object such that it was parallel to the eye level of a sitting participant. The object was placed at a 45 centimeter distance, as recommended by Hololens 2 intractable object guidelines with air tap interaction [16]. The size of the object was tested in the range between $1.4 \times 1.4cm$ and $3.5 \times 3.5cm$. Both participants felt comfortable with $1.4 \times 1.4cm$.

Event and task period. Each block in our experiments includes repetitive trials. If events are repeated too fast, task difficulty would increase, and task accuracy would suffer. If the events are far apart and the downtime between events could break the presence [1] We tested 1 to 15 second event periods; $< 3s$ periods were too small as participants reported being puzzled and unable to complete the task, participants felt bored for $> 8s$ periods. We chose 5 seconds as the period between events and tasks to keep the user comfortable while maintaining high task accuracy.

Cue appearance. We first tested the change in color as a visual cue for the participant in both sets of experiments (see details in Section 3.2). We found that the color change as a cue was distracting. After trying additional cues, we used objects appearing in the scene as a cue for the realistic vs. abstract. For the coffee mug experiment, we tested the glowing button as a cue for tasks, but the participants found it distracting. At the end, we used filling the coffee in the mug as a cue. The participant found that it was more engaging than the color change and the glowing button.

Proximity of interaction. We tested the air tap interaction at a close and far distance. The participant struggled with air tap. The reason for this could be that the virtual object was placed near the user and the far-touch interaction moved the virtual object farther from the participant and created unnecessary distraction. The participant found the near air tap interaction is more natural, so we kept it as the mode of interaction for all of our experiments.

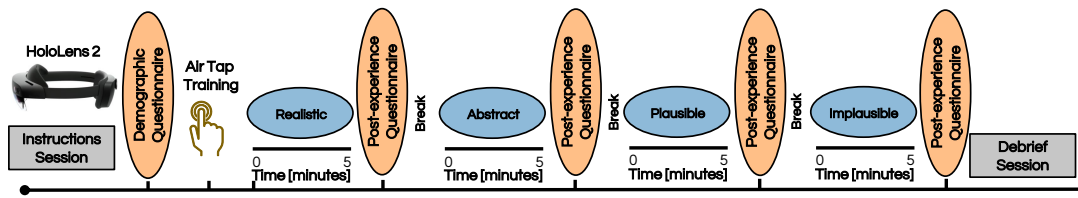


Fig. 2. Timeline for the user study consisting of pre-and post-questionnaire as well as four blocks across the two experiments.

Feedback. For realistic vs. abstract, we chose virtual object disappearing as a feedback for successful interaction, just like fruit ninja game. Participants found it acceptable. For plausible vs. implausible experiments, feedback was tricky without adding another virtual object in the environment. We tested with a subtle coffee-poured sound as feedback. One of the participants found it distracting. We then decided to use the mug color change as feedback rather than a cue and tested again. Both participants found that the color change feedback was more subtle and less distracting. Interestingly, the same "change in color effect" participant found it distracting when putting it as a cue instead of feedback.

4.5 Procedure

Figure 2 shows the complete timeline of the user study. After arriving at the study room, the experimenter welcomed the participants and gave them a verbal introduction and instructions. Participants then read and signed the IRB-approved informed consent form and filled out a pre-questionnaire about demographic characteristics (age, self-identified sex, and familiarity with the Mixed Reality (MR)). We also briefed them on MR and HoloLens 2). We then explain the flow of the study, the types of experiments, the duration of each block and experiment, their order, and the questionnaires. We further explained how to interact with the virtual objects, how to perform an air tap, and how hands should be held in their laps when not tapping. We also explain what to expect while wearing the headset, how and when they should interact with the virtual object, and what the cue would be for tapping and feedback.

As speeding could potentially override participants' natural instincts, we advised them on the importance of completing the task correctly, without the need to rush or stall. To avoid this instinctual bias, we deliberately kept this information (reaction time measurement) from the participant at the beginning and added this information in the debrief. Participants were also not informed of the frequency of tasks to minimize anticipation. We also used the talk-aloud protocol and asked participants to think aloud as they performed the tasks to assess the varied reactions of the participants overtime, which the subjective questionnaire does not capture. We also monitor the entire process through the first-person user view in Mixed Reality Capture [17]. Participants were also advised to stop at any point in the experiment if they felt uncomfortable.

We world-locked the virtual scene in the environment and put the headset on the participant. The reason to world-lock the scene is to remove environment variable, and world-lock is adjusted at participants' eye level by adjusting the chair of the participant. We adjusted the headset to the participant's head and calibrated it to the participant's interpupillary distance for the best visual results. Before the first experiment, for training purposes, participants were then asked to interact with the virtual object (an empty dialog box with the next button and cancel) using an air tap. The feedback was set to sound that both the experimenter and the participant could hear when the first air tap was successful.

We set the order of blocks for the participant as *realistic*, *abstract*, *plausible*, and finally *implausible*. In each block, the timer is started only when the participant is in a comfortable position and verbally confirms readiness. At the end of the 5 minute period, the block ends by closing the scene (through the device portal [18]). We then remove the headset from the participant's head and ask them to fill out three post-experience questionnaires (see more details in Section 3.3). After completing four blocks and post-experience questionnaires, we thanked the participants for their participation

No	realistic (μ , MAD)	abstract (μ , MAD)	F	p (<0.05)	plausible (μ , MAD)	implausible (μ , MAD)	F	p (<0.05)
ALL	4.97, 0.90	2.74, 0.74	107.63	yes	5.17, 0.93	2.81, 1.091	92.55	yes
SUS	4.92, 1.09	2.60, 1.89	70.89	yes	5.34, 1.09	2.70, 1.13	72.79	yes
IPQ	4.17, 1.58	2.31, 0.57	38.99	yes	4.51, 1.35	2.63, 0.90	38.56	yes
PQ	5.58, 0.65	3.09, 0.99	122.11	yes	5.61, 0.79	2.98, 1.30	86.51	yes

Table 1. Mean, mean absolute deviation (MAD), and F-test results for all questions and scores from the individual questionnaire.

and provided their compensation. The participants were then briefed on the study. After informing the participants about the purpose of the study, at this stage we provided them with the option to withdraw from the study.

5 RESULTS

In this section, we present and discuss the results quantifying *presence* using questionnaires and user *reaction time*. We also discuss the relationship between *presence* and *reaction time*.

5.1 Presence questionnaire scores

We collect answers to the questions in each of the three questionnaires on a 7-point scale. We obtain the *presence* score by averaging the 7-point scores of all the questionnaires individually and combined. We use non-parametric analysis of variance (ANOVA) tests to compute statistical results, since all questionnaires use an ordinal scale, which means that the answers are in order, but without a degree of difference between choices. We use Python's Scipy library to do all the calculations [49]. In both experiments, we changed the feelings of *presence* by manipulating the appearance of the object (i.e. *place illusion*) and the behavior of the object (i.e. *plausibility illusion*).

5.1.1 Overall and individual questionnaire scores. The results for both experiments are reported in Table 1. We found that manipulating the object appearance in the first experiment had a significant impact on *presence*. We observed $F(1, 38) = 107.63, p < 0.001$ for all the questions, $F(1, 38) = 70.89, p < 0.001$ for SUS, $F(1, 38) = 38.99, p < 0.001$ for IPQ, and $F(1, 38) = 122.11, p < 0.001$ and PQ questionnaires¹. Similarly, we found that altering the object behavior also had a significant effect on the overall *presence* scores, $F(1, 38) = 92.55, p < 0.001$, SUS ($F(1, 38) = 72.79, p < 0.001$), IPQ ($F(1, 38) = 38.56, p < 0.001$), and PQ ($F(1, 38) = 86.51, p < 0.001$).

Our results reveal that the experiment design that leveraged manipulating the appearance and behavior of the object was successful in altering *presence*, as there was a significant change in the mean scores of *presence*. The histogram of the scores across the users and its probability distribution is shown in Figure 3. The results also show that we satisfied the *sensitivity* criteria outlined in §3 that required a distinguishable difference between the sense of *presence*.

5.1.2 Questionnaire scores variance. We take the absolute difference from the mean as a measure of variance, mentioned as Mean Absolute Difference (MAD) in the Table 1 that presents the values across all the questionnaires. We use F-test to check whether the variance measures across the blocks are equal. A validation of null hypothesis will mean that questionnaire scores before and after our manipulations are spread equally from their respective means.

Our F-test scores for the appearance manipulation experiments showed that the null hypothesis cannot be rejected, $F(1, 38) = 0.980, p = 0.11$, and the variance of the scores did not change. Similarly, the F-test scores for the behavior manipulation experiments also showed that the variance of the scores did not change before and after the manipulation,

¹ $F(X, Y)$: X and Y are the degrees of freedom between and within the groups, respectively. $X = \text{no. of groups} - 1$, $Y = \text{size of the group} - \text{no. of groups}$.

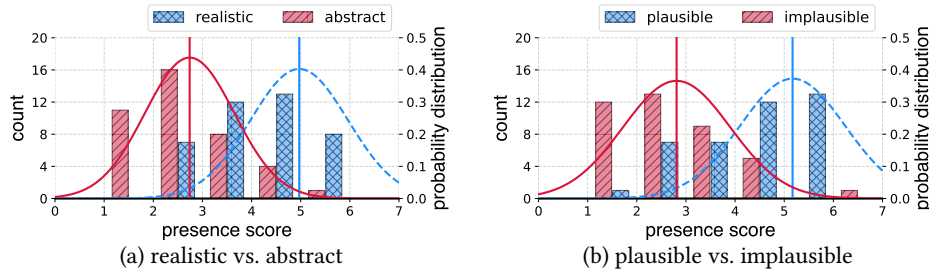


Fig. 3. Histogram, means (left y-axis), and fitted Gaussian distribution (right y-axis) using the average score across all questionnaires.

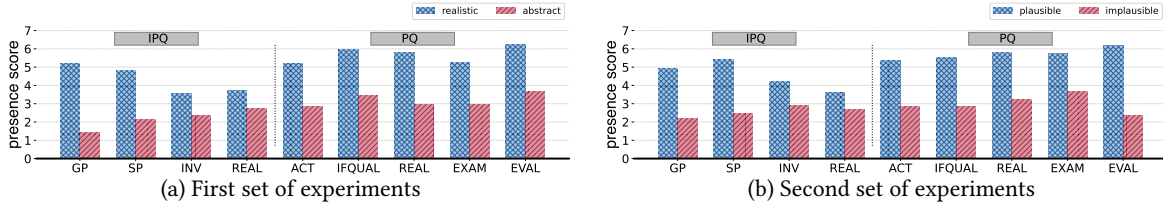


Fig. 4. Questionnaire Subscale Scores. SUS does not have any subscale. IPQ has general presence (GP), spatial presence (SP), involvement (INV), and realism (REAL); PQ has possibility to act (ACT), interface quality (IFQUAL), realism (REAL), possibility to examine (EXAM), and self-evaluation of performance (EVAL).

$F(1, 38) = 1.250, p = 0.64$. The statistics of the three individual questionnaires also showed that the variance measure did not change. The questionnaire-specific results are omitted for brevity.

5.1.3 Subscales. The IPQ and PQ questionnaires use subscales that break down the overall score into different components and allow for a more nuanced analysis of presence. The mean scores for all subscales are shown in Figure 4. Across IPQ subscales, in both sets of experiments, general presence (GP) and spatial presence (SP) are the two most significantly changed aspects of the questionnaire. The user involvement (INV) and realism (REAL) also changed, but not to the same degree. Across all subscales, we observed at least 1-point difference in scores on a 7-point score scale. Across the PQ subscales, no single subscale had a significantly higher change than the other subscales. These results indicate that our manipulation mechanism, despite being simple, impacted the sense of *presence* across all subscales of the questionnaires but to slightly different degrees.

5.1.4 Additional statistical analyses. We also measured the correlation coefficient between overall scores across all questionnaires before and after manipulation for both sets of experiments. We found that the scores are correlated with the 0.53 and 0.48 values for the appearance manipulation and behavior manipulation experiments, respectively. We also conducted paired t-tests for the two blocks in each experiment. We found the t-statistics values of 19.44 and 11.74 for the realistic vs. abstract and plausible vs. implausible blocks, respectively. In both cases, the p-value was significantly less than 0.001. Given this result and the significant difference in the mean of the blocks, we can say that the users who felt a greater presence before the manipulation were also the ones who felt a low presence after the manipulation. A similar analysis for the IPQ questionnaire suggested an even higher correlation of 0.67 and 0.57 for the first and second sets of experiments, respectively. PQ questionnaire showed similar results, which are omitted for brevity.

5.1.5 User characteristic and presence. We performed a regression analysis using F-test to check if age, gender, and familiarity with the MR had any impact on the *presence* scores. In this analysis, the null hypothesis is that a

Experiment	Reaction Time (ms) (μ MAD)	Change (%)	No. of Interactions
realistic	954 170		50
abstract	1313 290	37.63	44
plausible	930 240		54
implausible	1182 210	27.10	51

Table 2. Average, mean absolute difference (MAD), %age change of user reaction times, and average number of interactions across the two sets of experiments.

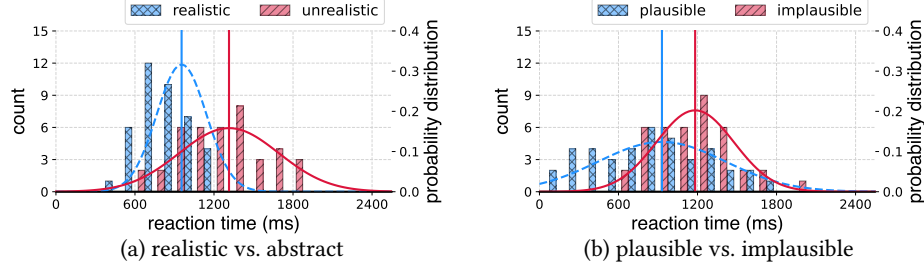


Fig. 5. Histogram, means (left y-axis), and fitted Gaussian distribution (right y-axis) using the average reaction time across participants. regression model based on a given variable is not a better fit than a simple intercept-only model. We observed that age, gender, and familiarity with MR did not have any effect on overall *presence* scores, as we obtained values of $F(1, 38) = 0.9561, p = 0.3412$, $F(1, 38) = 1.6370, p = 0.96$, and $F(1, 38) = 2.192, p = 0.7132$, respectively.

5.2 Reaction Time

Next, we evaluated the user *reaction time* across the different experiments. As mentioned above, we collect the time user **interacts** in response to *cue* and uses the difference between two timestamps as *reaction time*. Since our experiments run for at least 5 minutes, we get more than 60 data points per user per experiment. We use Python's Scipy library to do all the calculations [49]. Table 2 presents the high-level results of the experiments.

5.2.1 Overall reaction time scores. Our results show that our manipulations had a significant impact on the average user reaction times, which increased from 954 ms to 1313 ms and 930 ms to 1182 ms for the object *appearance* and *behavior* manipulations, respectively. This represents a significant change, 37.63% for appearance manipulations and 27.10% for behavior manipulations, in the user reaction time across experiments. Furthermore, as the users became more familiar with the environment and setup, the reaction times from one set of experiments to the other experiment decreased. However, this was not able to overcome the increase in reaction times due to our manipulations within a set of experiments. This is one of the reasons why we decided to order the experiments so that the manipulated experiment was conducted after the normal experiment. Furthermore, we also measured the number of times a user was able to respond to the cue. Since the number of cues between experiments differed due to slight variations in the experiment duration, we only considered the first 60 cues to collect these statistics. We see that the number of interactions within a given set of experiments decreased after the manipulations. This is because the user's overall presence and ability to act, as illustrated by the PQ subscale *ACT*, were reduced. We also see an improvement in the number of interactions across the two sets, as was observed for the improvement in reaction times.

5.2.2 Reaction time mean and variance distribution. Figure 5 shows the distribution of average reaction time scores across the users. We analyze how mean and variance compare between the two sets of experiments. The F-test

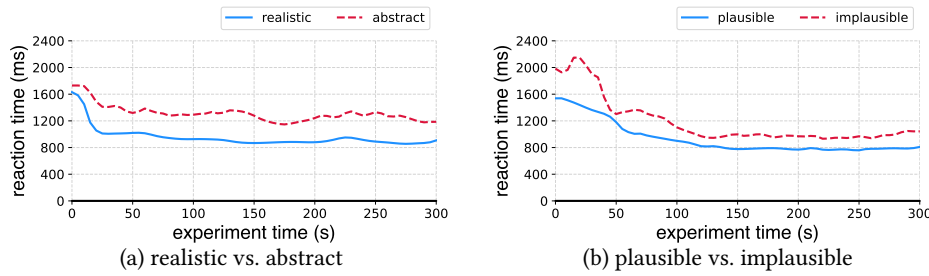


Fig. 6. Average user *reaction time* for different experimental settings. User reaction time improves over time (*recovery effect*), an effect that impacts all experiments in a similar way.

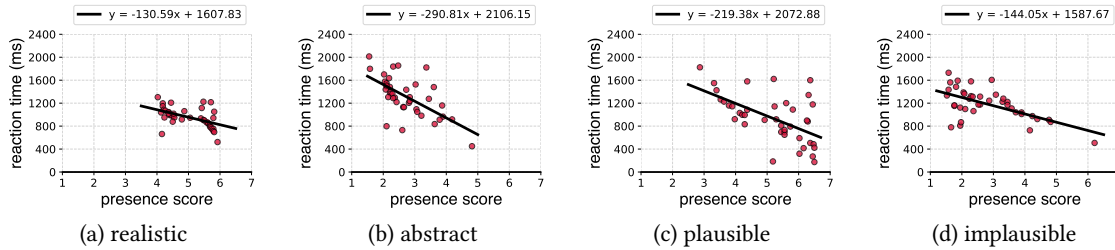


Fig. 7. Reaction Time vs. Presence: Reaction time decreases as the presence increases. Reaction time and presence also show modest correlation: overall (-0.65), realistic (-0.51), abstract (-0.63), plausible (-0.57), and implausible (-0.59). Each red circle represents a study participant. Black line is the linear regression fit for the data.

scores, $F(1, 38) = 35.74, p < 0.001$, for the mean user reaction time between realistic and abstract experiments suggested that the two samples are equally distributed from their means. The F-score for the plausible vs. implausible experiment was higher, $F(1, 38) = 9.77, p = 0.002$, but still statistically significant enough to prove that the participants showed very similar behavior in terms of their distribution of their means. Additionally, we also performed the variance test across experiments using the absolute difference of the mean as a measure of variance. The results showed that the variance measure did not change, as we obtained $F(1, 38) = 12.96, p = 0.0005$ and $F(1, 38) = 7.04, p = 0.0009$ for the realistic vs. abstract and plausible vs. implausible experiments, respectively.

5.2.3 Reaction time recovery over time. As the user gets familiar with the environment, the reaction time should improve. To analyze this phenomenon, we looked at the aggregate reaction time over time across users, as illustrated in Figure 6. We see that participants took significantly longer to respond to the cues at the start of the experiment. The reaction time drastically dropped and settled to a steady-state within the first 30 seconds for the realistic vs. abstract experiment (a). After that, there was only a modest improvement in reaction time over the duration of the experiment. The participants took slightly longer than 60 seconds to reach a steady-state for the plausible vs. implausible experiment. In addition to recovery over time, this result also highlights another aspect of user reaction times. First, the difference between the reaction times for both sets of experiments remains almost constant over time across the experiments. This means that the recovery effect is similar for the experiments with and without manipulation.

5.3 Presence - Reaction Time Relationship

In previous sections, we analyzed how our manipulations impacted the *presence* and *reaction time* of the study participants. We observed that as the average *presence* across all users decreased from realistic (4.97) to abstract (2.74) and plausible (5.17) to implausible (2.81) experiments, the user *reaction time* increased from 954ms to 1313ms and 930ms to 1182ms,

respectively. This is a significant result and suggests that *presence* impacts *reaction time*, at least in the confines of our study. To extract further proof of this relationship, we study the correlation between the individual *presence* scores for the study participants and their *reaction times*. Figure 7 shows the scatter plots of participants' *presence* scores and their *reaction times* for all the experiments. We observe that *reaction time* has an inverse correlation with *presence* score, albeit with different slopes for the linear regression fitted model across experiments. The correlation coefficients lie in the range of -0.51 and -0.63, suggesting a statistically significant negative correlation between the two variables. The overall correlation coefficient of -0.65 suggests a general negative correlation between *presence* and *reaction time*. However, the individual differences between different experiments are significant enough to warrant that the use of this relationship should be scenario specific. This is because a given change in *presence* score or *reaction time* does not illicit the exact same change in the other variable in all the experiments.

6 LIMITATIONS AND FUTURE WORK

For this study, we used two scenes consisting of a simple one-object scenario, a design choice to minimize the effect of variables other than *presence* on the reaction time. Therefore, the setup may not be sensitive to the broader effects of varying the feelings of *presence* on reaction time in a multi-object virtual scene. Future work should investigate how *presence* relates to reaction time with different degrees of scene complexity, cognitive load, and dynamic physical environment. Similarly, we investigated the effects of varying the feelings of *presence* with only periodic tasks and active interaction (tap). It is worth investigating how the *presence* and reaction time can be affected in other types of non-periodic tasks and passive interaction (eye gaze). Furthermore, participants reported their *presence* levels using the questionnaires that leverage Likert scales. Other questionnaires that rely on different assessment mechanisms such as open-ended questions might reveal additional insights into the results. We also acknowledge that our proposed approach depends on user interactions to measure the reaction time. In virtual scenes with little or no interaction, our technique may fail to produce any measurement. In future work, the measurement of reaction time should consider using some other backup mechanism, such as eye-gaze tracking, to combat low-interaction scenarios.

We have not found any effect of gender, age, and familiarity with MR on the *presence* or reaction time in this study. This can be due to our purposefully simple design, but in a complex scenario with some applications, these user characteristics may have an effect on the *presence* or reaction time. In addition, we have not tested the effect of break in the *presence*, but it may have some impact on the *presence* and reaction time. In the future, the break in the *presence* can be tested with the reaction time as a measure and may add to the discussion of the *presence*-reaction-time relationship. In this study, we did not ask participants to fill out a cybersickness questionnaire, However, during the study with participants, we asked participants to report any discomfort they feel during or at the end of the experiments. None of the participants reported any feeling of discomfort. This may be due to the limited exposure time and a wider adjustable range of interpupillary distance that Hololens 2 offers [15]. However, in the future, this variable can be tested in isolation to isolate the potential impact of cybersickness in varying the feelings of *presence*.

Despite the limitations of the work and the opportunity for improvements, we argue that our results present sufficient evidence of a relationship between *presence* and *reaction time* to justify a further discussion of whether a performance-based metric such as *reaction time* can be used to describe *presence*. Currently, post-experience questionnaires are the most commonly used measures of *presence* in previous work. However, a major disadvantage of such questionnaires is that they are based on the subjects' memories of the experience. Such memories can reflect an inconsistent and incomplete picture of the experience. *Reaction time*, on the other hand, is a passive and objective measure that does not depend on the comprehension of the user and the memory of the experience. In the future, with additional investigation,

it can serve as a measure of *presence* and as a feedback loop that developers can use to improve the run-time experience, since it measures the phenomenon at the moment when it is perceived.

7 RELATED WORK

Many studies have discussed the concept of presence in the context of a virtual environment [26, 46]. In 1996, Mel et al. conducted an experiment to assess the influence of presence on performance, while the participants learned to play three-dimensional chess [45]. They noted that immersion refers to a description of a technology and presence refers to concomitant behavioral and psychological responses of people.

Previous research highlights the importance of presence as an outcome of virtual environments [6, 27, 43, 50, 54] and multiple approaches such as subjective post-experience questionnaires [12, 38, 39, 46, 54], evaluating the behavioral and physiological response [11, 19, 20, 25, 30, 47] have been suggested to quantify presence. Sheridan [43] laid the foundation for determining the underlying factors of presence, such as sensory information, sensor control, and motor control. Over time, previous work developed, refined, and validated questionnaires to measure presence. Earliest approaches to measuring presence by varying the update rate of a computer-generated simulation on the sense of presence in a virtual environment [2]. They measured the fidelity of the interaction using two questionnaires. Wittmer and Singer [54] developed a 32-item presence questionnaire to test involvement/control, natural, and interface quality. Although the questionnaire is the most cited questionnaire, it is still criticized for subjectively defined factors. More questionnaires were developed that more or less measured aspects of realism relating to the virtual scene [12, 33, 36, 38, 39, 44].

However, the questionnaires cannot measure the time-varying qualities of the presence, can produce unstable, inconsistent, and unrepeatable responses, and are prone to user bias. Behavioral and physiological measures offer relief from user bias but can also often be noisy and unreliable [30]. Integrating questionnaires directly into the virtual experience has been explored to live assess virtual experience [40, 42]. Both studies did not have a clear baseline. Later, Schwind et al. used integrated questionnaires to measure presence. Similar studies were conducted to assess the break in the presence [1, 35]. The study asked participants to complete questionnaires that added to break-in presence and were prone to participants bias. Therefore, we make the case for measuring feelings of presence with an approach that is objective, reproducible, and produces a stable response without creating interference in the virtual scene.

8 CONCLUSION

We presented a user study (N=40) to understand the relationship between *presence* and *reaction time*. We changed the sense of *presence* of the participants by manipulating *appearance* (*place illusion*) and the non-task-relevant *behavior* (*plausibility illusion*) of the virtual object, and systematically measured the *reaction time* of the participants in response to visual stimulus. Our post-experience questionnaires show a significant change in the *presence* across experiments. Similarly, we see a significant change in user *reaction time* as we vary feelings for *presence*. Our analysis shows a negative correlation between the presence and *reaction time*. Furthermore, our study provides insight into the considerations for using *reaction time* as a possible measure of *presence*, as well as preliminary recommendations on the possibilities of future research to better understand the relationship between *presence* and *reaction time*. We found that as the average *presence* score for the two illusions decreased from 4.97 to 2.74 and 5.17 to 2.81 (on a 7-point scale), the average *reaction time* increased by 37.63% and 27.10%, respectively. Although our study points to an inverse relationship between *presence* and *reaction time*, we also note that the slope of the negative correlation may depend on the scenario under consideration and needs further investigation.

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